

TurboPress: What Each Ingredient of Rotated Low-Bit LLM Weight Quantization Actually Buys

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Abstract

Modern post-training quantization (PTQ) of large language model weights stacks several ingredients—random rotation, strong codebooks, Hessian error feedback, and mixed precision. We introduce **TurboPress**, a controlled harness that measures the *marginal* value of each ingredient at exactly matched bit budgets, and report four findings. (1) A corrected *equilibration exponent*: in the weight-only, post-rotation regime the optimal per-channel fold is a quarter power $s_j = (\mathbb{E}[x_j^2]/\|\mathbf{W}_{:,j}\|^2)^{1/4}$, distinct from the square root of AWQ/SmoothQuant and from the weight-activation balance of BASE-Q. (2) GPTQ/LDLQ error feedback in the rotated basis cuts 2-bit KL divergence by $5.0\times$ over nearest rounding at $\sim 3\times$ lower cost than trellis coding, and 3-bit error feedback beats nearest 4-bit. (3) Trellis coding and error feedback are complementary: trellis wins at 2 bits, both are near-lossless at 4 bits. (4) KL-sensitivity-driven mixed precision beats uniform precision at an equal 3.0-bit budget. We also present a rigorous *negative result*: unbiased (QJL) residual correction, though provably debiasing, loses to deterministic quantization for weights. We deliberately avoid a state-of-the-art claim and quantify the scale and baseline gaps that separate these measurements from one.

1 Introduction

The strongest PTQ pipelines share a backbone: incoherence processing [2, 3], a strong quantizer [12, 13], and Hessian-aware error feedback [4]. Rather than add a frontier method, we ask: *at matched bit budgets, how much does each component buy, and why?* TurboPress is a harness in which each component can be toggled and measured against a full-precision reference by KL divergence $\text{KL}(p_{\text{fp}}\|p_q)$, top-1 next-token agreement, and perplexity, with storage reported to the exact fractional bit. Our claims and the evidence for each are mapped in Table 1.

Table 1: Claims and supporting evidence.

Claim	Evidence
Optimal rotated weight-only equilibration is a quarter power.	Prop. 1; §2
Error feedback $\gg$ nearest, complementary to trellis.	Table 2; Fig. 1
Larger models quantize better at low bits.	Fig. 3
KL-driven mixed precision beats uniform at equal budget.	Table 3
Unbiased QJL correction for weights is counter-productive.	§3

2 Method

A layer $\mathbf{y} = \mathbf{W}\mathbf{x}$ is factored exactly as $\mathbf{y} = (\mathbf{W}\mathbf{D}\mathbf{R}^\top)(\mathbf{R}\mathbf{D}^{-1}\mathbf{x})$, with $\mathbf{D} = \text{diag}(\mathbf{s})$ an activation-aware equilibration and $\mathbf{R}$ a seeded randomized orthogonal transform; we quantize $\mathbf{W}'' = \mathbf{W}\mathbf{D}\mathbf{R}^\top$.

Equilibration (quarter power). The rotation spreads the per-row quantization error isotropically, so the expected output error factorizes.

Proposition 1. *With $c_j = \|\mathbf{W}_{:,j}\|^2$, $m_j = \mathbb{E}[x_j^2]$, and normalized quantizer $MSE \varepsilon^2$, $\mathbb{E}\|\Delta\mathbf{y}\|^2 \approx (\varepsilon^2/d)(\sum_j c_j s_j^2)(\sum_j m_j/s_j^2)$, minimized (Cauchy–Schwarz) at $s_j = (m_j/c_j)^{1/4}$.*

The AWQ/SmoothQuant fold $s_j \propto \mathbb{E}[x_j^2]^{1/2}$ is optimal only for column-aligned error; the rotation breaks that alignment, shifting the optimum to a quarter power. This differs from BASE-Q [5], whose 1/2 balances *activation* against *weight* quantization noise. The prediction $\alpha \approx 1/4$ is consistent with the small AWQ exponents found by grid search [8] and the $\alpha=0.3$ frozen by QAM-W [11].

Quantizers and error feedback. We use rate- R trellis-coded quantization [9, 13] with data-free trellis-optimized Gaussian codebooks (the rotation makes the source distribution known), storing exactly R bits/weight. We also implement GPTQ/LDLQ [3, 4] in the rotated basis, forming the transformed Hessian $\mathbf{H}_z = \mathbf{R}\mathbf{D}^{-1}\mathbf{H}\mathbf{D}^{-1}\mathbf{R}^\top$ with the fast transform; it reduces to nearest rounding when $\mathbf{H}_z$ is diagonal.

Bit allocation. We measure each layer’s KL at each width by quantizing one layer at a time, then greedily allocate a fixed average budget to the upgrades with the best KL-drop-per-bit.

3 Experiments

We quantize all 196 decoder linears of Qwen3-0.6B [10] in fp16 on a single 8 GB GPU (embeddings/head/norms kept full precision), evaluating on 4,080 next-token positions of real text against the fp16 reference (perplexity 29.11); calibration uses a disjoint 2,048-token slice. SmolLM2-135M [6] is the scale-up reference.

Table 2: Qwen3-0.6B sweep; fp16 perplexity 29.11. Best per bit in **bold**.

Configuration	bits/w	KL	top-1	ppl	time (s)
Nearest 2b	2.012	10.282	0.020	564,874	4.6
Nearest 3b	3.013	1.641	0.381	138.66	4.4
Nearest 4b	4.013	0.399	0.670	43.91	4.0
TCQ+eq 2b	2.023	1.215	0.450	91.42	245.9
TCQ+eq 3b	3.023	0.289	0.710	38.58	245.2
TCQ+eq 4b	4.023	0.079	0.843	30.63	247.8
GPTQ/LDLQ+eq 2b	2.023	2.072	0.383	177.10	79.0
GPTQ/LDLQ+eq 3b	3.023	0.312	0.707	36.28	88.8
GPTQ/LDLQ+eq 4b	4.023	0.095	0.844	30.45	77.2

Error feedback vs. trellis. Error feedback cuts 2-bit KL $10.28 \rightarrow 2.07$ ($5.0\times$) and 3-bit $1.64 \rightarrow 0.31$ ($5.3\times$) at $\sim 3\times$ lower cost than TCQ; GPTQ/LDLQ 3b beats nearest 4b. TCQ wins at 2 bits (KL 1.22 vs. 2.07); the methods tie at 3–4 bits—evidence they are complementary.

Exponent ablation. Fig. 2 tests Prop. 1 directly: the same rotated scalar quantizer with only the fold changed. The quarter power dominates at every bit-width on Qwen3-0.6B and Qwen3-1.7B; on the 1.7B model the AWQ-style square-root fold is *worse than no equilibration* at 3–4 bits (KL 3.58 vs. 2.24; 1.85 vs. 0.64) while the quarter fold improves every setting—the over-scaling failure the proposition predicts.

Scale-up. For TCQ+eq, the 2-bit perplexity ratio drops $8.3\times \rightarrow 3.1\times$ from 135M to 0.6B, and the 4-bit gap from +8.1% to +5.2% (4-bit KL is size-invariant at ≈ 0.079); see Fig. 3.

Per-layer allocation. MLP projections dominate sensitivity; query/key are nearly free (Fig. 4). At an equal 3.0-bit budget the allocated model beats uniform 3-bit (KL 0.238 vs. 0.276, ppl 34.76 vs. 36.59; Table 3).

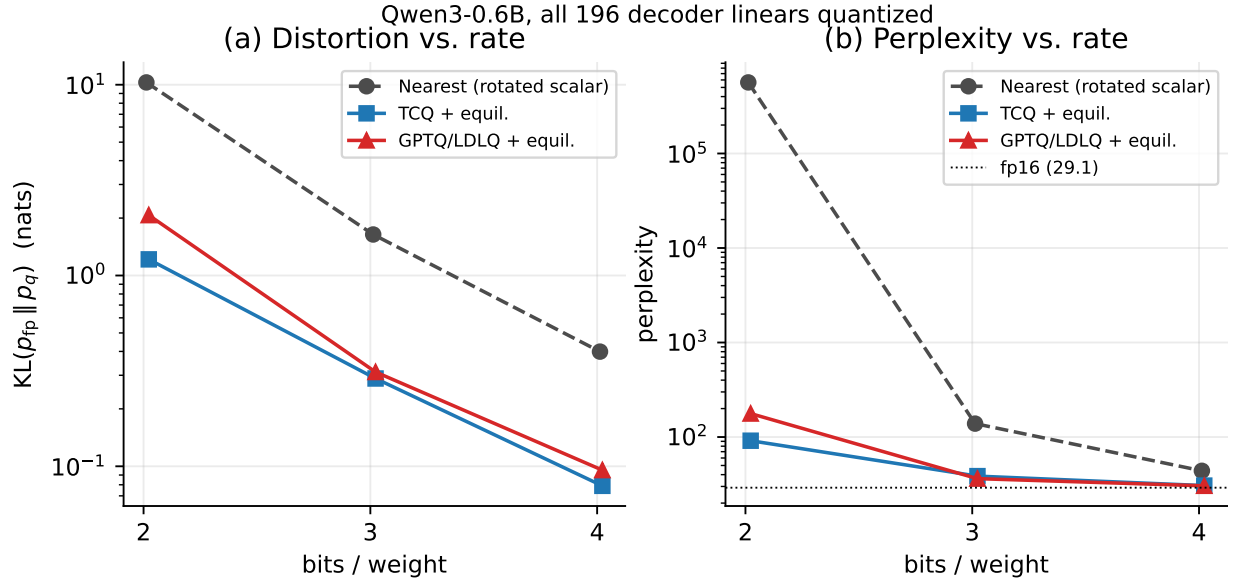


Figure 1: Rate-distortion on Qwen3-0.6B. Trellis coding and error feedback both dominate nearest rounding and are near-lossless at 4 bits.

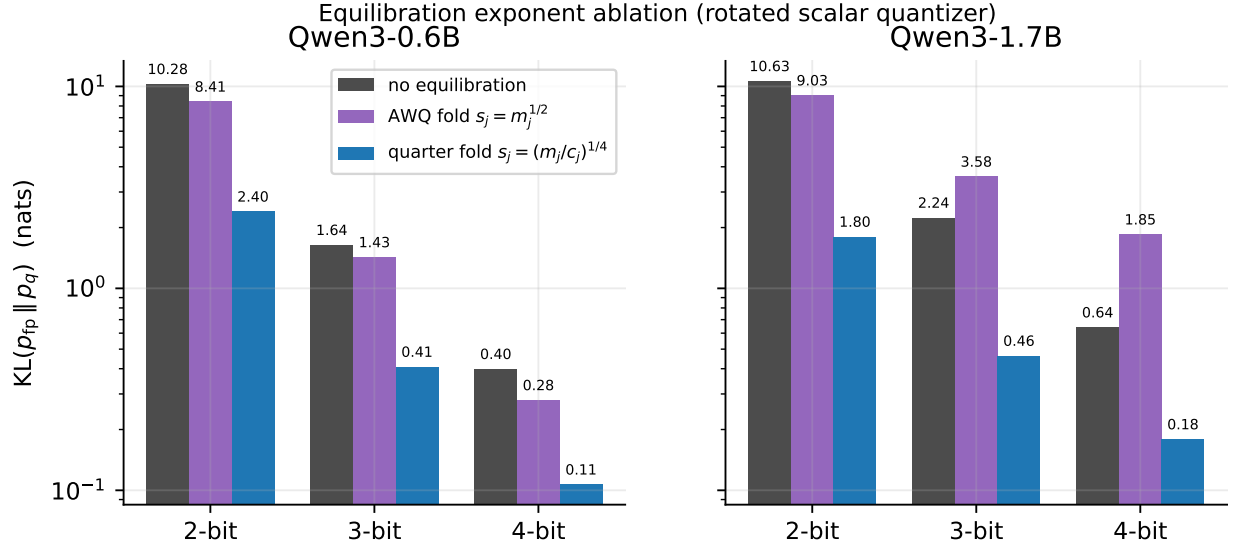


Figure 2: Equilibration-exponent ablation: identical rotated scalar quantizer, only the per-channel fold differs (log scale, disjoint calibration/eval).

Table 3: Allocation vs. uniform (TCQ+eq). The 3.0-bit row is equal-budget.

Budget	Model	bits/w	KL	ppl	layers (2/3/4)
3.0 (equal)	Mixed	3.023	0.238	34.76	24/149/23
	Uniform 3b	3.023	0.276	36.59	0/196/0
2.5	Mixed	2.523	0.494	44.79	101/91/4
–	Uniform 2b	2.023	1.386	105.18	196/0/0

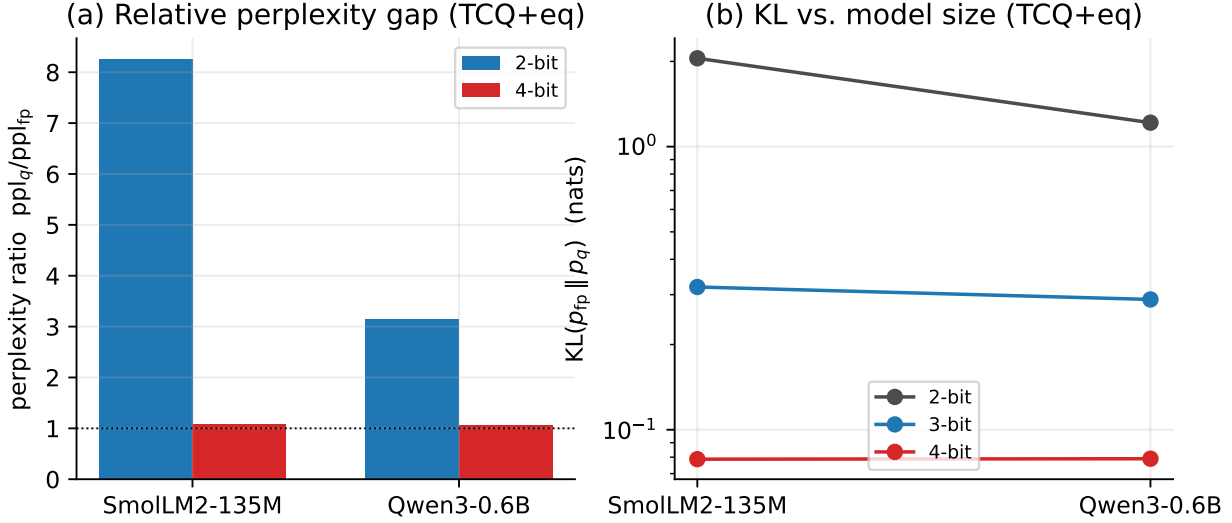


Figure 3: Scale-up 135M→0.6B (TCQ+eq): the aggressive low-bit settings improve most.

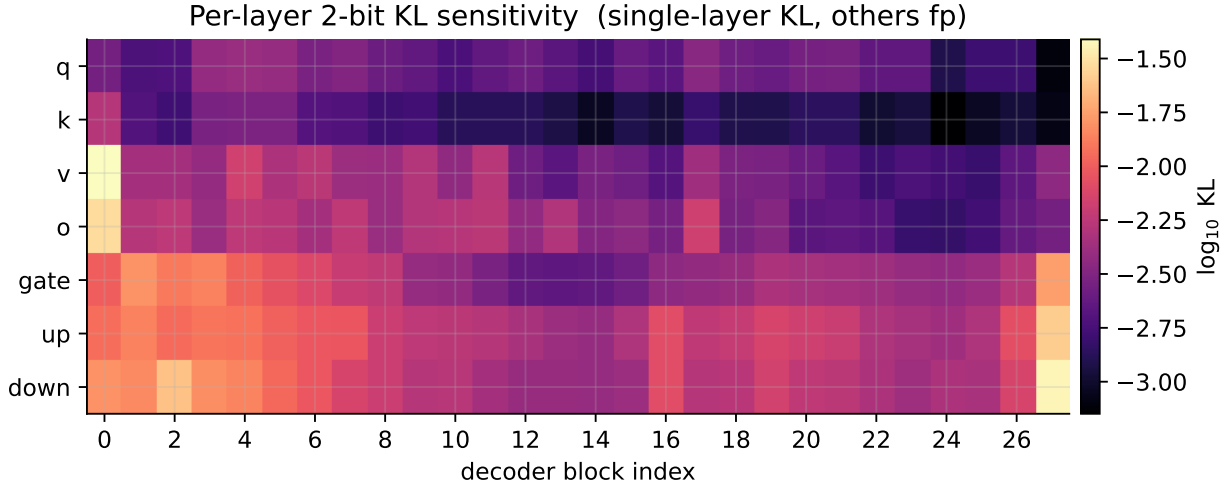


Figure 4: Per-layer 2-bit KL sensitivity map ($\log_{10}$). MLP and value/output projections carry the sensitivity.

QJL-for-weights negative result. At equal ~ 3 -bit budget, unbiased QJL correction is far worse than nearest 3-bit: on a 16-layer synthetic chain, KL 2.58 vs. 0.096; on real SmolLM2-135M, KL 9.75 vs. 1.72 ($5.7\times$). The estimator debiases exactly, but its variance ($\sqrt{\pi/2} \|r\| \|x\| / \sqrt{k}$) exceeds the removed bias even at a full extra bit, and compounds through depth; the rotation already decorrelates residuals from activation structure. QJL remains appropriate for shallow KV-cache inner products [14].

4 Related Work and Positioning

The pipeline shape overlaps recent work [1, 5, 7, 11–13, 15]; we claim no state of the art. Our defensible contributions are the quarter-power equilibration exponent (Prop. 1)—positioned against BASE-Q’s $1/2$ and AWQ’s grid-tuned heuristic—and the QJL negative result. The AM–GM / Cauchy–Schwarz machinery for optimal scaling is itself established [1, 5]; only the exponent for the weight-only, post-rotation regime appears to be unstated elsewhere.

5 Conclusion

TurboPress measures, at matched budgets, what each ingredient of rotated PTQ buys. The results motivate combining error feedback with trellis coding (block-LDLQ over TCQ), evaluated at 7B against standard baselines—the natural methods follow-up.

Reproducibility Statement

All experiments are seeded and deterministic on a given device. The supplementary release contains: the quantization modules (rotation, scalar/trellis quantizers, LDLQ error feedback, equilibration, allocation); the sweep and allocation drivers with their exact command lines; 89 unit tests covering exact math identities, statistical unbiasedness/variance bounds, the trellis bit-stream round-trip, LDLQ=nearest under diagonal Hessian, and greedy-allocation properties; and the raw result JSON files from which every table and figure in this paper is generated by a single script. Models (Qwen3-0.6B, SmolLM2-135M) and the evaluation corpus are public. Proposition 1 is stated with its modeling assumptions in the main text.

Ethics Statement

This work concerns compression of existing, publicly released language models and introduces no new model capabilities, data collection, or human-subjects component. Quantization can shift a model’s output distribution; we therefore report a distributional metric (KL divergence) rather than perplexity alone, and we caution that low-bit quantization may amplify existing model biases or degrade safety behaviors in ways not captured by perplexity—an evaluation we did not perform and flag as necessary before any deployment. We foresee no additional dual-use risk beyond that of the underlying models.

References

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A Paper Checklist

1. **Claims match contributions?** Yes. Scope and non-claims (no SOTA) are stated in the abstract, Table 1, and Related Work.
2. **Limitations discussed?** Yes: scale ($\leq 0.6B$), lack of released-baseline / WikiText-2 / zero-shot comparison, and the modeling assumptions behind Prop. 1.
3. **Theory: assumptions and proofs?** Prop. 1 states its isotropic-error and MSE-proportional-to-energy assumptions and gives a proof sketch; it is an approximation, not an exact optimality theorem.
4. **Experimental reproducibility?** Yes: seeded code, exact configs, public models/corpus, and result JSONs are released; figures regenerate from JSON.
5. **Data and code released?** Code, tests, and result logs are in the supplement; models and corpus are public.
6. **Compute reported?** Single 8 GB consumer GPU; per-configuration quantization wall-clock is in Table 2 (4–248 s).
7. **Metrics and error reporting?** KL divergence, top-1 agreement, and perplexity over 4,080 fixed positions; storage as exact fractional bits/weight. We do not report variance across seeds—a limitation.
8. **Existing assets credited?** Models, corpus, and all baseline methods are cited.
9. **Broader impacts / ethics?** See the Ethics Statement.
10. **Safety of released artifacts?** The release is quantization code and numeric logs; no models or data are newly released.